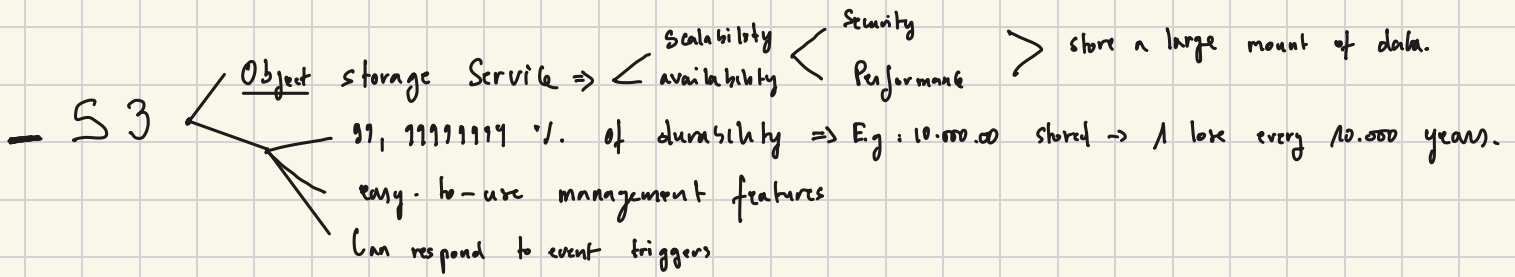
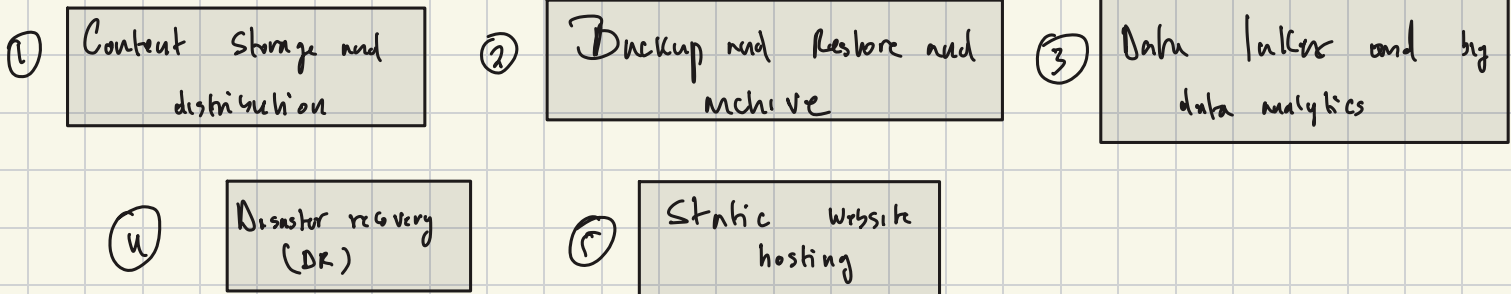




# AWS S3



## Use Case



- 1 Securely store static content and distribute them from S3 or CloudFront.
- 2 Backup that your data and use storage lifecycle policies to archive rarely accessed data.
- 3 Centralized repository that allow to store structured and unstructured data at any scale.
- 4 Protect critical data, app and IT systems with cross-region replication and asynchronous copying of obj. across buckets.

## S3 Components

1. **Bucket** : `https://s3-<aws-region>.amazonaws.com/<bucket-name>/`
2. **object** : `https://s3-<aws-region>.amazonaws.com/<bucket-name>/<object-key>`

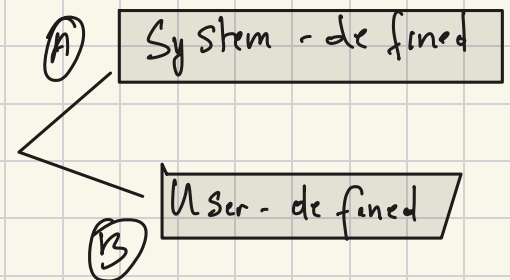
# + Creation of S3 buckets and Working

1. Must Specified a region.
2. Consider: Latency, Cost.
3. Can change region after creation.
4. bucket name must be unique.

Structure: `Https://<bucket-name>.s3-<aws-region>.amazonaws.com/object-key`

→ Use prefixes to imply a folder structure in an S3 bucket

→ S3 buckets has **Metadata**



## (A) S.D

- Information that S3 Controls

- \* object creation date
- \* object size
- \* object version

- Information you can Modify

- \* Storage-Class Config.
- \* Server-Side Encryption

## (B)

### U.D

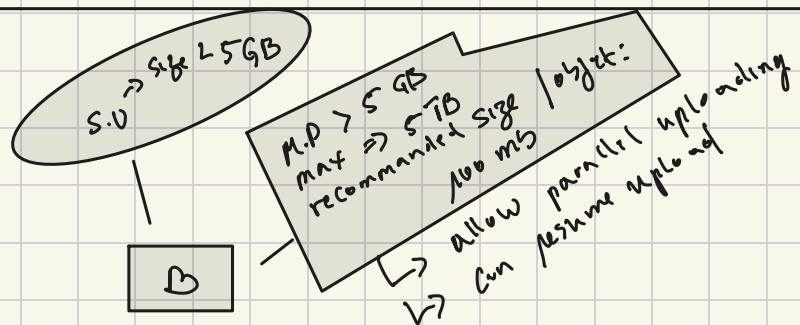
- Infos that you assign to the object

**Syntax** ⇒ X-ami-z-meta

## 1. Put object

⇒ Upload Programmatically to S3 bucket

↳ Single and multipart



2. Get ⇒ retrieve a complete object < Complete a range of byte: for large object.

3. ! Select operation.

Versioning: It's a way to keep multiple variants of an object in the same bucket

- In versioning-enabled S3 buckets, each object has V. ID
- When versioning is enabled  $\Rightarrow$  Can't be disabled but Suspended

Delete operations

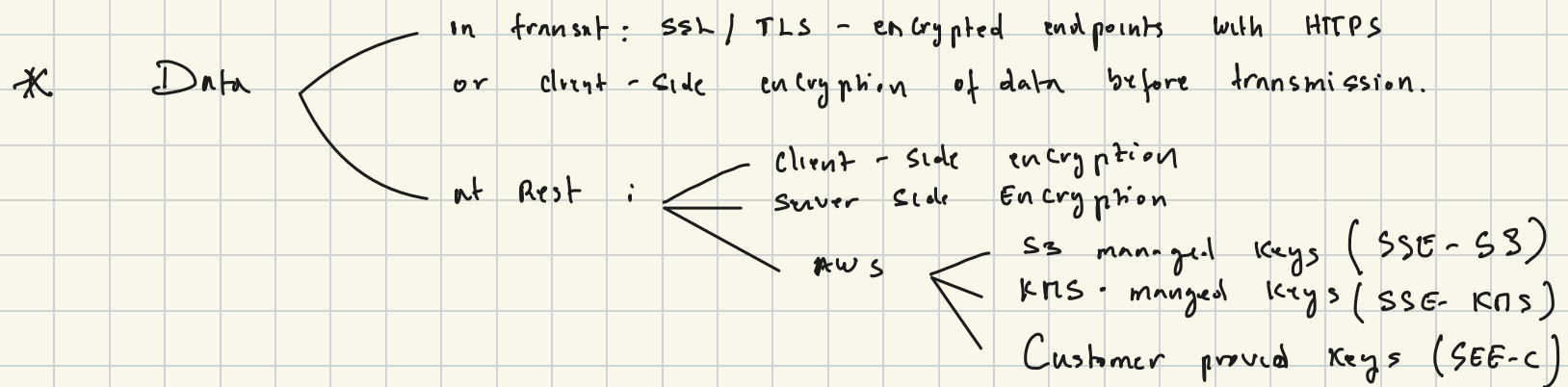
Versioning disabled  $\Rightarrow$  Delete permanently the objects.

Versioning enable  $\Rightarrow$  Delete ops Insert a delete Marker into the bucket.  $\Rightarrow$  Get Ops  $\Rightarrow$  404

$\hookrightarrow$  delete a Specific Version  $\leftarrow$  obj key-value  
version ID.

$\Rightarrow$  If you want to delete permanently  $\Rightarrow$  Delete each Version of the object.

Protecting data and Manage  
access into S3



$\Rightarrow$  S3 will only encrypts objects that are added to the bucket after encryption is enabled

Security of the objects  
 $\Rightarrow$  data

## => Security of the bucket

□ Identity-based policies: granted permissions to users.

→ admin; users or groups => attached to an Identity

□ Resource-based policies: apply to resources  $\leftarrow$  <sup>S3 buckets</sup> objects  $\rightarrow$  are the R

→ different type  $\leftarrow$  <sup>object ACLs</sup>  
Bucket policies  
Bucket ACL

ACL: access control list: manage access at the object level or bucket level

=> Use to grant basic write/Read permissions to other AWS account

bucket policies => Manage Cross account access to S3 buckets.

## => Presigned URLs

↳ Provide access to put or Get object, without granting permission to perform any action

↳ Use the permission of the user who creates the URL

↳ provide Security credentials, a bucket name; an object key and Http method and an Expiration date and time.

↳ only valid until the Expiration time (Max of 1 week)

## => CORS: Cross origin resource Sharing.

→ Must be allow to accept req from other domain. => Can be specified for certain domain only => only them can access.

